

Automatic Zoom

INTELLIGENT DETECTION AND CLASSIFICATION OF AERIAL OBJECTS USING DEEP LEARNING

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Abstract

This project presents an extensive investigation into the development, training, evaluation, and

deployment

of a deep-learning-based aerial object classification system capable of distinguishing between

drones, birds,

planes, and miscellaneous airborne entities. As global airspace becomes increasingly saturated

with

commercial, private, and autonomous aerial vehicles, ensuring safety and situational awareness

has never

been more critical. Conventional detection systems such as radar and acoustic sensors continue to

struggle

with low-RCS (radar cross-section) targets such as consumer drones, which fly at low altitudes and

often

share similar signatures with birds. Consequently, computer-vision-based detection has emerged

as a

promising alternative approach.

This dissertation expands existing research by developing a fully reproducible framework that

incorporates

modern CNN and transformer-based architectures, rigorously evaluated across multiple dimensions

including classification accuracy, inference latency, generalisation, dataset imbalance, robustness

to

noise/environmental distortion, and interpretability through saliency mapping. The project uniquely

compares baseline custom CNN architectures with state-of-the-art pretrained models such as

ResNet-50,

MobileNetV3-Large, and ConvNeXt-Tiny, evaluating both frozen and fine-tuned variants under

identical

conditions. Additional focus is placed on the engineering challenges associated with dataset

preparation,

including the construction of a unified aerial-object dataset from multiple sources, implementation of

preprocessing pipelines, augmentation strategies, and automated hyperparameter tuning.

The resulting system demonstrates that fine-tuned ConvNeXt-Tiny delivers exceptional accuracy,

achieving

over 96% performance on the test set, while MobileNetV3-Large demonstrates outstanding

inference speed